

Enabling Continuous, In-Field Introspection: The Programmable IPU Architecture

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Abstract—Modern silicon systems are increasingly opaque, leaving developers blind to in-field hardware behavior on real workloads. Existing introspection tools force a debilitating trade-off: fixed-function Performance Monitoring Units (PMUs) are low-overhead but lack programmability, while trace-based solutions are high-fidelity but incur prohibitive data and performance overhead, and have an accessibility challenge as well. This leaves a critical “capability gap,” with no solution offering programmability, low overhead, and real-time analysis simultaneously. This paper introduces the Introspection Processing Unit (IPU), a new, lightweight architectural primitive designed to fill this gap. The IPU is a novel hybrid microarchitecture, pairing a programmable RISC-V core with judiciously chosen microarchitectural accelerator extensions, including a tiny eFPGA. This hybrid design is our core insight, balancing the flexibility of software with the line-rate efficiency required for non-invasive introspection. We demonstrate the IPU’s power not through point solutions, but by showing its ability to perform classes of analysis previously considered intractable in the field. We show a single IPU architecture can perform: 1) Stateful Emulation: (e.g., A/B testing instruction prefetchers), 2) Software-Defined Performance Attribution (e.g., creating per-instruction cycle stacks), 3) Scalable, On-Chip Data Aggregation (e.g., fine-grained GPU utilization monitoring), and 4) Real-Time Component-Level Diagnosis (e.g., per-miss root-cause diagnosis of a prefetcher). Our complete RTL-level prototype, synthesized at 7nm, proves this new capability is practical, with an area overhead of $< 1\%$ and power consumption of $< 25mW$.

Index Terms—introspection, analysis, accelerator, A/B testing, performance monitoring

I. INTRODUCTION

As modern CPUs, GPUs, and accelerators grow increasingly complex and opaque, our ability to understand their *after*-deployment behavior remains archaic, creating a crisis in observability. Developers, researchers, and performance analysts are fundamentally blind to the detailed, real-time operation of hardware in the field, forcing them to rely on simulation, proxies, and offline analysis that cannot capture the intricacies of live workloads. The rich data made available from live collection and analysis, in tandem with advanced generative AI frameworks, could make major advancements in computer hardware feasible at a much quicker rate than is currently possible [64]. This blindness to live data is not for a lack of tools, but for a lack of the *right* tool. The existing landscape for hardware introspection forces a debilitating trade-off between flexibility, efficiency, and accessibility:

1) **Fixed-Function Monitors (e.g., PMUs)**: Performance Monitoring Units (PMUs) are the workhorse of in-field analysis. They are ubiquitous and impose near-zero overhead. Their fundamental limitation, however, is their inflexibility. A PMU can only count what its designers anticipated would be useful. It is a rigid, fixed-function tool that cannot be adapted to answer new questions that arise post-deployment, and hides nuances of actual on-chip events [34].

2) **High-Overhead Trace (e.g., JTAG)**: At the other extreme, trace-based solutions provide exhaustive, cycle-by-cycle detail. While invaluable for post-mortem debugging in a *lab*, they are entirely unsuitable for in-field, real-time analysis. The resulting data deluge (often terabytes per second) overwhelms on-chip buffers, saturates I/O bandwidth, and induces significant application slowdowns, making them unusable for observing long-running, production workloads. JTAG is impractical for a software developer seeking insights on bottlenecks on the hardware they are optimizing code for.

3) **Specialized Hardware (e.g., TEA for PICS [33], [34])**: To address specific, critical needs like **detailed program analysis**, designers have proposed specialized, fixed-function hardware blocks to obtain per-instruction cycle stacks (PICS). They can offer deep insights for programmers such as which instruction spent how many cumulative cycles in L2 vs. TLB misses. While effective for their narrow task, this approach is unscalable. It is not feasible to anticipate and ship dozens of disparate hardware monitors for every *potential* use case.

This leaves a vast and critical **capability gap**. There is no existing architectural solution that provides **programmability** (like a CPU), **real-time signal access** (like a trace), and **low overhead** (like a PMU) simultaneously. We lack the ability to deploy new, sophisticated analysis *software* that runs directly on the hardware, observing its internal state, aggregating data locally, and even participating in low-latency tracking, all while the main system runs at full speed. Section VII presents a taxonomy of the relevant academic literature as well.

This paper proposes the **Introspection Processing Unit (IPU)**, a new, lightweight architectural primitive designed to fill this gap. It is distinguished by three key properties:

- **Low-level Signal Access**: The IPU has high-bandwidth, read-only access to a curated set of internal microarchitectural signals, far beyond what is exposed to a PMU.

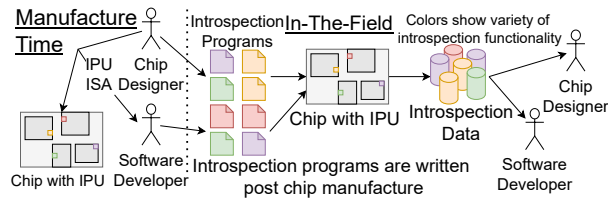


Fig. 1: IPU overview

- **On-Chip Data Processing:** It executes analysis programs *locally*, enabling on-chip aggregation, filtering, and statistical analysis. This “computational introspection” solves the data-deluge problem by exporting only high-level insights, not raw traces.
- **Low-Latency State Tracking:** Its tight integration allows it to track system state in real-time, enabling tasks like hardware emulation that are impossible at line rate with any other tool.

Concretely, an IPU is built around a tiny RISC-V core placed in close physical proximity to the hardware block it introspects (a “Hardware-module Introspection Target” or HIT), as shown in Figure 1. At design time, engineers can attach IPU’s to HITs, exposing a set of curated internal signals (e.g., up to 32 per IPU). The actual analysis that runs on these signals is fully programmable via “introspection binaries” that can be securely conceived and deployed any time during the chip’s life. From a system-level perspective, the IPU streams its aggregated introspection output to the host via a logical FIFO, with chip designers using software signing to control access.

The core contribution of this paper is the design, implementation, and evaluation of this introspection architecture that executes flexible, software-defined analysis. We also propose a security model to govern its use via a trusted computing base and signed introspection binaries, addressing the challenges of this new capability. To demonstrate the power and generality of *this single architecture*, we illustrate its application to four disparate capabilities, each representing a class of problem currently considered intractable in the field:

- 1) **Stateful Emulation:** We implement the complex entangled instruction prefetcher [63], demonstrating the IPU’s ability to run stateful simulations *while* the processor runs at full speed.
- 2) **Software-Defined Performance Attribution:** We use the IPU to replicate in software the functionality of a specialized PICS block, providing per-instruction cycle stack accounting for program optimization.
- 3) **Scalable, On-Chip Data Aggregation:** We use IPU’s to monitor GPU TensorCore utilization, performing local data aggregation to provide system-wide insights.
- 4) **Real-Time Component-Level Diagnosis:** We attach an IPU to a state-of-the-art data cache prefetcher [14], classifying demand misses into root-cause categories by tapping prefetcher-internal decision-point signals.

We present a complete RTL-level prototype, synthesized at

7nm, which proves the IPU’s practicality. Our design achieves this new, powerful capability with an area overhead of $< 1\%$ and a power consumption of $< 25\text{mW}$, making it a feasible addition to any modern chip design. The IPU provides a path forward, moving hardware introspection from a rigid, pre-defined function to a flexible, programmable platform.

Contribution and Architectural Novelty. To deliver these capabilities, our IPU architecture and microarchitecture are a hybrid system, not simply a standard RISC-V CPU. This design is the result of several key insights, unknown prior to our demonstration, that were necessary to achieve low-overhead, high-performance introspection. While we use a RISC-V core as the baseline for **Programmability**, a core contribution is the **judiciously chosen microarchitectural extensions** that enable high-bandwidth, allowing the core to subscribe to internal hardware signals. We recognized that a processor-core-only approach is insufficient for line-rate data processing, and this led to our central insight: a **hybrid core-accelerator architecture**. We pair the core with small, specialized hardware, including a “histogram accelerator” and a tiny, on-chip eFPGA (used when introspection is dominated by state-machine traversals). We demonstrate that for a wide class of introspection tasks, this minimal reconfigurable logic is sufficient. This hybrid design—balancing the flexibility of a CPU with the efficiency of purpose-built hardware—is the key to achieving the IPU’s capabilities, a novel “sweet spot” that prior work has not identified.

This paper is organized as follows. An overview of the IPU and system design is in § II, software and hardware architecture are detailed in § III and §IV respectively. Evaluation methodology is in § V, and our results demonstrating the four capabilities are in § VI. § VII covers related work.

II. SYSTEM OVERVIEW

This section provides an overview of the IPU’s hardware and software components, which are illustrated in Figure 2. We first define two key terms: the “user program” refers to an application running on the chip’s main processor (e.g., a web browser, DL inference workload). The “introspection binary” is the small, signed program that runs on the IPU core to analyze the user program’s hardware-level behavior. To contextualize this relationship, we outline two representative classes of introspection tasks. First, a task of **on-chip data aggregation**: an IPU monitoring a TensorCore’s output values every cycle. Instead of streaming these high-frequency values off-chip, which is infeasible, the introspection binary computes a histogram of the value distribution locally (e.g., in 1024 bins). It then sends only this small, final histogram to host memory, solving the data deluge problem. Second, a task of stateful, **low-latency emulation**: an introspection binary implementing an instruction prefetcher. The IPU subscribes to the CPU’s front-end signals (e.g., the current PC), runs its own prefetch logic in real-time, and reports the resulting miss-rate, enabling a kind of live A/B testing.

Hardware. The Introspection Processing Unit (IPU) is a modular design built around a simple in-order RISC-V core.

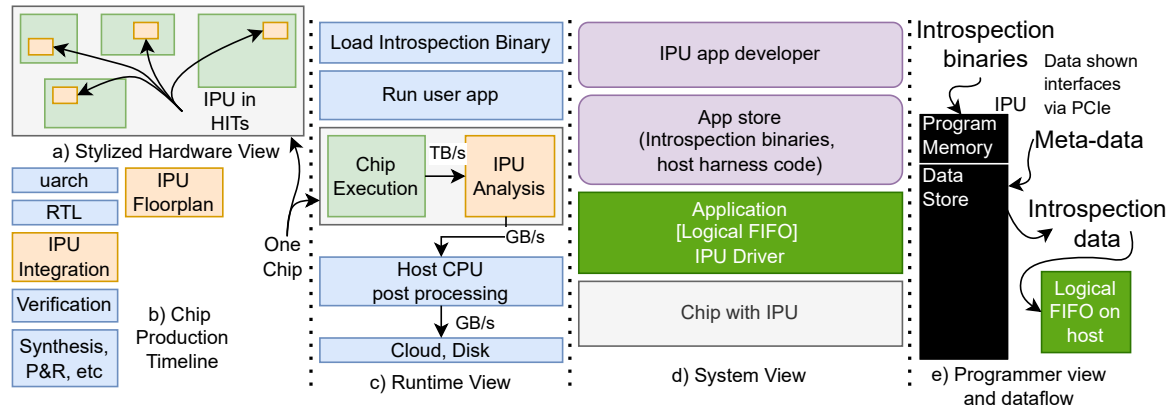


Fig. 2: Cloud is purple, chip/HIT is green, blue is existing processes, orange is our contribution/new circuitry

Inclusion of common analytics functions beneficial to maintain speed creates the IPU_{lite} . The IPU_{pro} has soft logic for finite state machine traversal introspection programs as we detail in §IV. An IPU is responsible for monitoring an individual hardware component of the underlying chip: the Hardware-module Introspection Target (HIT). Our design associates one IPU per HIT (Figure 2(a)). Examples of possible HITs include individual sub-modules within a core, L2 controller, GPU SM sub-modules, a CPU core, etc. The IPU will be flattened into the hierarchy of the HIT for placement and routing (P&R) purposes (Figure 2(b)). The IPUs are also connected to the underlying chip’s on-chip network (OCN), which is used to transfer introspection information out of the IPU.

Runtime View. The runtime view of a program is cascaded in Figure 2(c) with time flowing downward. The IPU driver exposes an API to configure and load the introspection code. The user application can specify what regions to analyze (e.g. specific tensor address regions). Once IPU configuration is complete, the user application begins execution. On each data input to the IPU, it determines if the data is within the region to analyze. If so, the introspection program executes on said data. While analysis is on-going, any further incoming data is dropped. Therefore, introspection programmers must be cognizant of the data rate specifications from the hardware designer. Once analysis execution completes, the IPU awaits the next data. In most introspection programs, outputs occur at intervals through a logical FIFO (Figure 2(e)). Once the user application ends, the IPU output can be post processed and, finally, exported onto disk or into the cloud.

System Architecture. The system architecture (Figure 2(d)) includes an API to expose IPUs and the IPUs themselves. At runtime, the IPU appears as a PCIe device that is programmed via an API, allowing hardware and software developers to download introspection programs. Each IPU is physically tied to its hardware block, and secure code execution is enforced via certificate-based code-signing using a public/private key infrastructure similar to Android/iOS app stores [1], [4]. This enables a secure, app-store-like model where trusted introspection programs—authored by chip designers, researchers,

or developers—are deployed through configuration API calls.

Privacy. Figure 2(e) illustrates the IPU’s security implications. By design, an IPU cannot inject signals into its HIT; it accepts code-signed binaries and emits introspection data. Rich analysis inherently risks leaking microarchitecture details—a known issue in tools like PICS [34] and even performance counters [26]. The central privacy concern is unintended inference, such as users learning undisclosed HIT details. We mitigate this with three policy modes enforced during code signing: **closed** (no third-party introspection), **restrictive** (requiring source-code review), and **permissive** (for credentialed developers). Future formal guarantees may leverage program verification [49], trace wringing [21], local differential privacy [15], [18], [28], [31], [75], or secure multiparty computation [44]. While subtler side-channel risks exist, the IPU’s risk is comparable to existing performance counters. Our system-level API further ensures user privacy by streaming only introspection data, without host state access (e.g., IP/MAC addresses). The broader concern of vendor-user trust is not unique to IPUs; diagnostics (e.g., JTAG) are often disabled in untrusted settings. Like confidential modes in modern CPUs/GPUs [58] or signed Intel firmware, IPUs can support SKU variants or boot-time configurations that disable them entirely. *In untrusted deployments, IPUs would be disabled, aligning with industry norms and not detracting from their value in trusted or OEM-controlled contexts.*

Limitations and Future Work. A potential critique is that the IPU merely shifts the design-time burden from event selection (PMUs) to signal selection. We argue this mischaracterizes the contribution: a PMU hardwires a fixed question (e.g., count L2 misses), whereas the IPU designer selects primitive data sources (e.g., L2 request bus, PC address). The IPU’s **Programmability** then allows forming a combinatorial number of new questions from these signals, including those the designer never anticipated. The IPU thus provides an extensible substrate for analysis, not a rigid, predefined answer.

A second concern is the potential for new side-channel attacks. We acknowledge this risk, which is inherent to any powerful introspection mechanism, including even simple per-

formance counters [26]. However, as detailed in our security overview (Figure 2), the IPU’s architecture provides a more robust and controllable security model than the status quo. While formal guarantees remain an area for future work, the IPU provides a securable foundation for introspection.

Additionally, the IPU is scoped to high-performance processors where its area is negligible relative to its HIT; embedded systems are left for future work.

III. SOFTWARE ARCHITECTURE

This section describes the software implementation on top of IPU’s with a deep dive on introspection programs. The collection of IPU’s on chip appear as a single PCIe device with distinct memory mapped areas for each IPU. The IPU’s also expose a host API for configuring what introspection binary to run and the trigger logic.

A. Programmer’s Model

An IPU is a programmable RISC-V core with a unique interface: 32 named inputs per execution. API calls configure which regions of the user program are analyzed. A small on-IPU memory is available, and the introspection program runs once per input set, repeating if new data arrives or idling otherwise. Post-processing can be performed at program end before offloading results.

Data & Transfer Semantics. A HIT’s signal interface to these 32 named values would be exposed to users through a formal documentation such as an ABI Spec specifying the semantics of signals and data arrival rate. Introspection developers must ensure that the HIT signal rate matches data processing speed in the IPU’s, use sampling, or be aware that some data will be dropped if there is a mismatch (See also §III-C). Introspection binaries are able to transmit data via host-memory mapped into the IPU’s address space. The IPU can then issue simple memory instructions to the unit’s memory hierarchy that are then transparently routed to a region in host memory. Using this, we can create a store for introspection results.

Small input buffers can delay when data is dropped; however, they cannot prevent it if the introspection rate is slower than the data arrival rate—by Little’s Law, loss is inevitable without stalling the HIT, which our architecture prohibits. Thus, the IPU drops inputs when active. Introspection programs must be designed with this in mind, using sampling, aggregation, or exploiting event sparsity. (See also end of §IV). Though the IPU runs at 2 GHz in 7 nm, HITs may run faster (e.g., 3–4 GHz). We do not require clock synchronization; instead, we support three modes: (1) Accept reduced fidelity via sub-sampling (e.g., 1-in-2 cycles); (2) Use a fast buffer for asynchronous, windowed processing; or (3) Restrict introspection to low-frequency phases or optimize the IPU for speed.

B. Software API and Execution Management

For configuration, the IPU exposes a minimal host-side API facilitated via memory-mapped I/O to the IPU. To configure a binary for execution, `IPU_CONFIG_IMAGE(image)`

specifies that an IPU should load a given introspection binary image. To configure when analysis begins, `IPU_CONFIG_START(addr)` sets the IPU to begin processing new data when the program to be analyzed reaches `addr`. `IPU_CONFIG_STOP(addr)` similarly sets when to stop processing new data. To enable fine-grained analysis we expose `IPU_PAUSE()` and `IPU_RESUME()` to pause and resume introspection execution. Finally, the command `IPU_FINALIZE()` instructs the IPU to execute any clean-up code necessary for the introspection program. The API resembles CUPTI [57] or Linux’s `perf_event` [29], allowing either a harness process to transparently configure and trigger IPU execution, or user binaries to use `PAUSE/RESUME` for fine-grained control.

C. Development Environment and Validation.

Introspection programs are written in C using a standard RISC-V toolchain. The IPU’s specialized instructions — histogram accumulation, hash computation, loop directives — are exposed as compiler intrinsics and a function library, so developers need not write assembly directly. The code snippets in this work show assembly to illustrate low-level behavior, but the actual development experience is C-level. For *IPU_{pro}* programs that include eFPGA logic, the developer provides the corresponding Verilog alongside the C control code.

To validate an introspection program before deployment, we provide an IPU emulator comprising a RISC-V functional emulator and, for *IPU_{pro}*, a Verilog co-simulator for the eFPGA. The emulator is coupled with a HIT traffic injector: a configurable testbench that generates signal streams at realistic timing and data rates derived from the ABI spec. Under any deployment policy mode, the developer has access to the ABI spec providing all the signal details needed to develop and test introspection code. Because the traffic injector is timing-coupled to the emulator, it faithfully reproduces the IPU’s dropping behavior (because IPU is an in-order processor, timing simulation is easy and fast). Developers observe exactly which inputs are dropped under realistic conditions, measure the resulting approximation error, and can iterate: tuning sampling window lengths, adjusting analysis granularity, and verifying that qualitative results (e.g., PC rankings in PICS) remain stable. Professional embedded workflows for microcontrollers routinely use traffic-injecting testbenches to validate firmware against realistic stimulus prior to deployment; using such a simulate-then-deploy (for example hobbyist platforms like Arduino [6] and the industry standard [5]). The IPU development flow follows the same paradigm, scaled to the demands of microarchitectural introspection.

D. IPU Program Lifetime

This section is an end-to-end example. The HIT is a CPU core and introspection development is closed policy - the CPU core designer will also write the introspection code. The analysis is PICS generation; the demonstration of the software-defined performance attribution capability (§VI-B) - we encourage the reader to skim that first.

Signal	#Bits	Reg	Semantics
itlb-miss	1	x0	Instruction TLB miss flag
icache-miss	1	x1	L1 Icache miss flag
recycle	1	x9	Recycle ROB unique IDs flag
fetch-pc-head	64	x11	Next PC to be fetched

TABLE I: Partial ABI Spec.

Pre-Fabrication. As part of the design of the CPU core, up to 32 signals are chosen to connect to the IPU. No information needs to be released to the public because of the closed policy; yet, an internal ABI Spec would be created to facilitate development. A partial ABI Spec is seen in Table I. Prior to verification, the IPU is flattened into the core layout and the HIT-IPU connections are made as outlined in Figure 2. A subtle issue is that the HIT designer needs to determine what the important signals are. By providing 32 we give them freedom to be profligate which allows rich analytics post-manufacture. Optimizations for future work include multiplexing a larger number of signals from a HIT to the IPU to sidestep the 32 signal limit. It is also possible for designers to create IPU variants, each with a different set of signal inputs. The type of IPU variant integrated into a particular HIT could then be chosen at random. If produced at scale, this would allow greater flexibility without generating larger overheads or compromising analytics accuracy.

Development. With the ABI Spec defined, the introspection code can be developed, which is the PICS generation in this example. To this end, the CPU designer references the ABI spec for each of the 17 signals necessary and identifies which input registers they are connected to. A portion of the compiled code handling the instruction TLB miss event is shown below:

```
_main: regtimer 50000, psv_loop
psv_loop: beq x0, 1, itlbm_m
beq x1, 1, icache_miss; x1 is a HW inp sig
...
itlb_m: hash r1, x12; x12 is an HW inp sig
ld r2, r1, 0
addi r2, r2, 0x40
sw r2, r1, 0
ret
...
```

In development, a 400,000 cycles sample rate is chosen to limit the output load; this creates approximation error that the CPU designer tests and finds within acceptable limits. The overall PC ordering by most cycles used is correct. The CPU designer releases the introspection binary, output location, approximation error, and if a region of interest can be chosen onto the app-store.

Analysis. Now, a SW developer has encountered unexpected slowdowns in their application and wants to profile it. They can download the PICS generation binary on the host. The listing also indicates that output will be put in a file on disk and that the user can optionally specify a region of instructions to analyze. They include a few API calls at the top of their program source:

```
IPU_CONFIG_IMAGE("PICS-generation")
IPU_CONFIG_START(ROI_BEGIN)
IPU_CONFIG_STOP(ROI_END)
```

PC	Event Combination	Number of Cycles
0x7912d0	DTLB miss, DCache Miss	5000000
0x80dda0	Branch Mispredict	200000
...		

TABLE II: Some rows from PICS generation analysis output.

<program code>

ROI_BEGIN and ROI_END are the beginning and end of the region of interest where the developer believes the slowdown to be. Essentially, it sets the respective Program Counters as the address to monitor for activating the IPU.

Post-Analysis. The output file has a list formatted as in Table II that is produced by host code, monitoring the IPU's introspection program. The developer uses the results for application performance tuning.

IV. HARDWARE ARCHITECTURE

A chip can have multiple IPU. Each IPU observes hardware signals from its corresponding HIT and runs introspection binaries on those signals as stylized in Figure 3(a). The IPU uses the chip's OCN, which enables them to transmit introspection outputs to and configuration from the host. The collection of IPU is visible as a PCIe device and uses PCIe to interface with the host; each IPU is distinguished through distinct memory mapped regions. If the chip lacks an OCN, some form of network which connects the IPU to the PCIe interface must be added as well.

Configuration of an IPU occurs through an API call:

IPU_CONFIG_IMAGE(image). This bundles the introspection binary and trigger logic meta-data and sends them to the appropriate IPU based on a given hardware device ID.

Interface overheads. HIT-IPU connections are short because the IPU is flattened into the HIT during P&R. Therefore, wiring overheads are negligible. Depending on the signals in a HIT and what the HIT itself is - there could be timing issues that can be addressed with standard buffering techniques used for performance counters. Consider a square HIT that is $2mm^2$. At a simple level, signals might need to traverse $2.8mm$ (half the perimeter of the HIT) to reach the IPU logic. To avoid timing issues for a high frequency design, one flip-flop might be necessary. Since this "far away" signal is buffered by one cycle, it implies all signals of this $HIT \rightarrow IPU$ must be buffered. HIT designers can use P&R feedback to judiciously select signals to avoid/minimize this.

No cross-chip wires are introduced; IPU configuration and output transfers use the existing OCN and PCIe, transmitting small packets at long intervals with negligible traffic impact.

Signal Selection Methodology. A natural question is how a hardware designer should choose which signals to connect to an IPU, and whether 32 signals are sufficient. We address both concerns. Across our four capability demonstrations, the maximum number of signals required was 17 (for PICS generation), while the prefetcher emulation and GPU aggregation demonstrations required only 2 and 3 signals respectively. These demonstrations span fundamentally different analysis

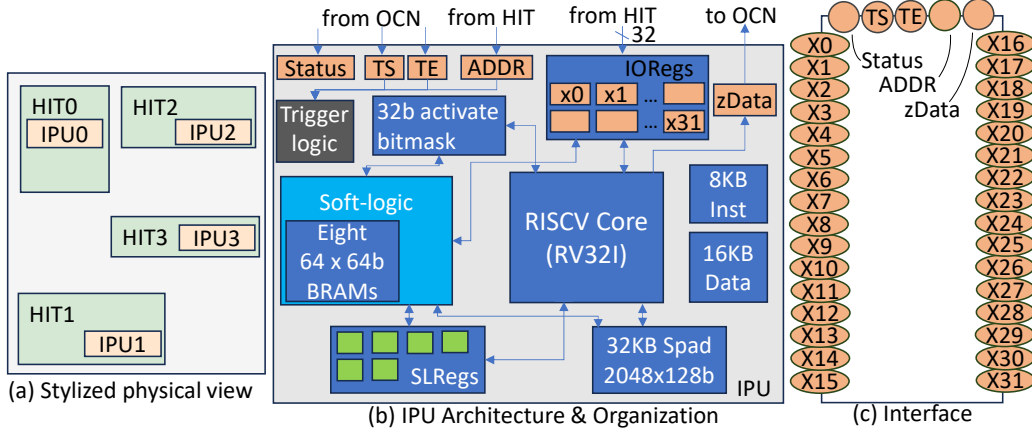


Fig. 3: IPU hardware architecture.

classes, yet none approached the 32-signal budget, suggesting it is well-sized for practical introspection tasks.

We observe that signal selection is principally determined by the HIT type rather than by any particular downstream analysis. A CPU front-end block naturally exposes fetch-related signals (the current PC, miss indicators); a cache controller naturally exposes its request bus, eviction activity, and coherence state; a GPU scoreboard naturally exposes functional unit activity flags. This property is important: the designer does not need to anticipate future analyses. Instead, they expose the important internal state, and the IPU’s programmability enables a combinatorial space of analyses to be composed from these primitives post-deployment.

This stands in sharp contrast to PMU design. A PMU hardwires both the data source and the analysis into a single counter (e.g., “count L2 misses”), producing exactly one pre-defined answer per event. The IPU decouples source selection from analysis: 32 primitive signals serve as inputs to arbitrary software, enabling analyses the designer never anticipated — all from the same fixed interface. Where N PMU events yield N answers, N IPU signals yield a vast space of programs.

To guide signal selection, we recommend a coverage-based methodology organized along three axes: (1) datapath values such as addresses and operand fields; (2) control signals such as stall conditions, flush events, valid bits, and branch outcomes; and (3) state indicators such as queue occupancy, buffer fullness, and functional unit activity. Ensuring representation across the major pipeline stages and known resource bottleneck points within each category provides a robust default signal set for a given HIT.

Hardware organization. The IPU architecture comprises four baseline components: a programmable core, a scratchpad SRAM, 32 input registers (IORegs), and 3 trigger registers: Trigger Start (TS), Trigger End (TE), and ADDR. The trigger logic looks at the ADDR signal from the HIT along with TS and TE (programmed using API calls) to control when the IPU becomes active. This is shown in Figure 3(b) while the interface is laid out in Figure 3(c). To concretize the

Unit	Area(mm ²)
RISC-V Core	0.0011
eFPGA	0.22
SRAM	0.016
Histogram Unit	0.001

TABLE III: Area Per-Component

architecture, we select some sizes for these components: the core is 32-bit RISC-V core (RV32I instruction set) using a data-SRAM and an instruction-SRAM; the scratchpad is 32KB.

Execution Model. An IPU has a 4-bit STATUS register, putting it in 6 states: PAUSED (P), ACTIVE-PAUSED (AP), ACTIVE-RUNNING (AR), FINALIZE (F), ERROR (E), and UNDEFINED (U). Its program structure includes 3 predefined program regions: `init`, `_main`, and `end`. Borrowing from the simplicity of micro-controllers, `init` is hardcoded to instruction memory address 0x0. The entire instruction memory is 8KB which amounts to 2048 instructions. On power-on, PC is set to 0 and starts executing the code in `init`. By convention, `finish` is hard-coded to be 16 instructions from the bottom of the instruction memory at 0x7F0. When the IPU is set to the finalize state, it executes code in the `finish` function and transitions to the PAUSED state.

The execution model of code on an IPU is data-driven, i.e. when new inputs arrive the `_main` function is called if the IPU is in the ACTIVE-PAUSED state. If it is running code triggered by previous input, it will be in the ACTIVE-RUNNING state - data received when in this state is dropped. Whenever we show a datapath of X bits for the IO registers, there is an implied additional valid bit associated. This bit is used to determine whether new input has arrived.

Microarchitecture. We design two IPU variants for different regions of the analysis-vs-data-rate design space. **IPU_{lite}** is a compact, cacheless RISC-V core with built-in primitives like histograms, loop counters, and hash functions for efficient, low-complexity introspection. The pipelined architecture of the **IPU_{lite}** is shown in Figure 4(a), while that of the included histogram unit is shown in 4(b). **IPU_{pro}** augments the RISC-V core with soft-logic—a lightweight embedded FPGA seen

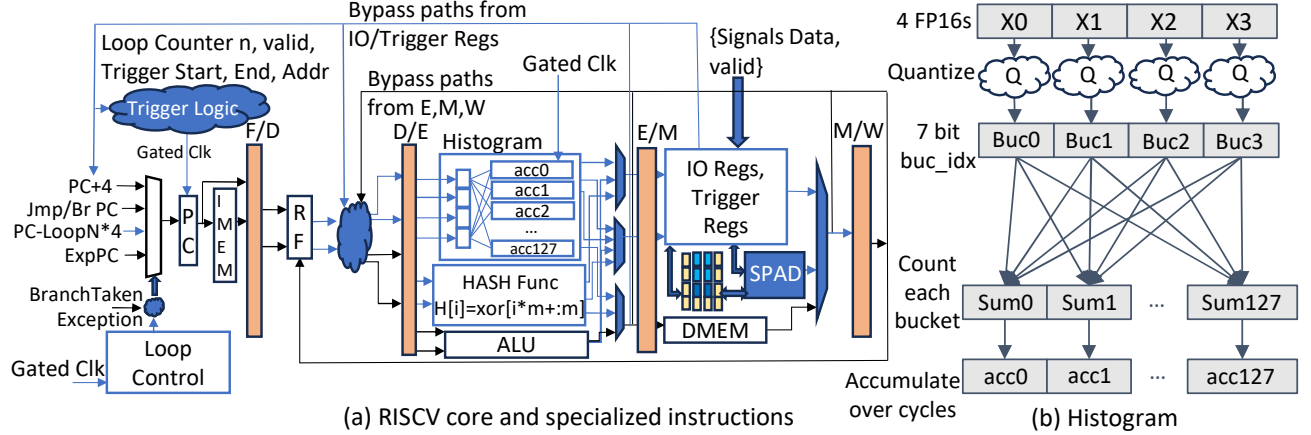


Fig. 4: IPU Microarchitecture showing datapath and control-path changes

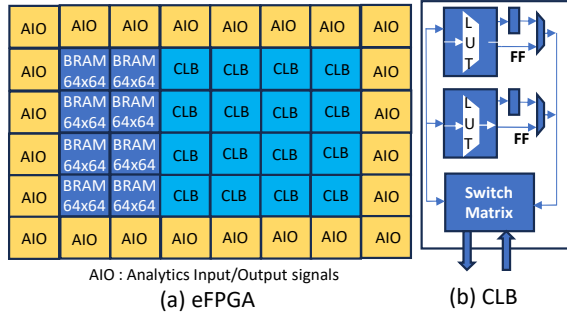


Fig. 5: IPU_{pro} soft-logic design

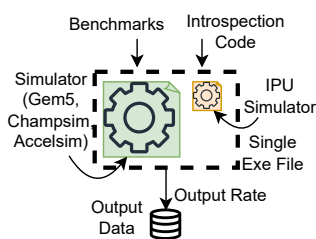
in Figure 5 with 590 configurable logic blocks (CLBs), 470 analytics IO (AIO) tiles (identical to baseline IO tiles except for the fact that we removed staging flip-flops, which are already in our IORegs), and eight small BRAMs (64 entries deep and 64-bits wide)—enabling complex, high-throughput analysis tailored at runtime. It interfaces via memory-mapped registers and supports introspection programs that bundle RTL logic alongside control code. Each IPU is embedded with its associated HIT and communicates via the OCN, avoiding long wires and limiting system traffic due to low data rates. Multiple IPU_s scale well: 5 IPU_{pro} and 10 IPU_{lite} consume just 0.65% of a 200 mm² die, and even full coverage across GPU SMs stays under 1% chip area overhead. A per-component breakdown of IPU area can be found in Table III.

On statistical bias from dropped data. When dropping data at a regular interval it is possible to introduce statistical bias. However, a simple algorithmic defense against pathological correlation is available entirely in software: randomizing the sampling window length around the target average (e.g., perturbing a 256-cycle window pseudorandomly). This breaks any systematic alignment between the IPU’s processing cadence and periodic hardware behavior, and requires no hardware changes which further illustrates the value of the IPU’s programmability.

V. EVALUATION METHODOLOGY

To empirically evaluate the IPU, we demonstrate three capabilities that we briefly describe in the introduction. Table IV describes the emulation and simulation testbeds we built. **Emulation and Simulation testbed.** Our four demonstrations span prefetch engine (Champsim [32] simulator), core-microarchitecture (GEM5 [8] cycle level simulation), and GPU cycle-level simulation (AccelSim [39]). We built an IPU emulator for code development and to determine correctness of the introspection. For performance (time), we developed a co-simulation environment that adds an IPU simulator into Champsim, Gem5, and Accelsim (left Figure in Table IV). For area and power, we implemented and synthesized RTL which was verified with an emulator for correctness. Table IV also shows the number of lines of code for the `_main` function of each introspection program. In total, more than 200 applications were simulated.

RTL Implementation. We implemented IPU_{pro} and IPU_{lite} in Verilog. Our implementation was verified for many input values against the introspection reference implementation. We use the AsAP7 7nm educational PDK [16]. For SRAMs we use CACTI scaled from 32nm to 7nm per [71]. We implemented our soft-logic using the FABulous design flow [40] to estimate area and power, and their synthesis flow for utilization. To determine soft-logic power, we used data from the reference introspection execution to create input traces. We used ASIC process flow of synthesis (DC Compiler/Priming), APR(Innovus), and VCD based power estimation obtained from Netlist simulation of all demonstrations. The max clock frequency for the soft-logic and IPU_{pro} is 1.3 GHz and 2 GHz for the IPU_{lite}. The HIT signals we need are described in each capability, and we show that the signals are readily available for any reasonable implementation of a GPU or core. The first two demonstrations are on a CPU and the third demonstration is on a GPU. For comparison, our references are: a CPU Zen2 4-Core Complex [70], which needs 31 mm² area and consumes 4 watts of power [3], and a GPU SM, that uses 3.475 mm² [46] of area and consumes 1 watt [38], [83].



Capability Demonstrated	Stateful Emulation	Software-Defined Performance Attribution	Scalable, On-Chip Data Aggregation	Real-Time Component-Level Diagnosis
Benchmarks	135 Traces [19]	13 Spec	21 Gemms	189 Traces [32]
Simulator Config Matching	Champsim Entangled [63]	Gem5 SE TEA [34]	AccelSim QV100 Model	Champsim Gaze [14]
#lines of IPU code	300 Verilog	75	2	50
#bits into IPU	64	215	4	132
#signals from HIT	2	17	3	6
Output size	3B	6B	4B	384B
Output timing (cycles)	Program approx 0	400k	256	20M
Rate per HIT (1/s)		15KB	15.6MB	18.75KB

TABLE IV: Methodology configurations and IPU System Testbed Flow for emulation in figure.

Full Synthesis and Place-and-Route. To complement our top-down comparison against published areas of modern high-end microprocessors, we performed full synthesis and place-and-route of the IPU integrated into a real processor. We selected BOOM (4 wide OOO core). We evaluated both a single-core configuration (with a 256KB L2 cache) and a 4-core complex (with a 1MB L2 cache, resembling the Zen2 4-core complex organization). The IPU was flattened into the core's hierarchy, with signals connected for the TEA case study (Section VI-B), and full place-and-route was performed using Cadence Innovus on the ASAP 7nm PDK. Timing was met in all configurations. To stress-test signal routing, we conservatively inserted 4 stages of flip-flop buffering on the HIT-to-IPU signal paths which models the wiring distance of a core far larger than BOOM even though our place-and-route results showed that only 1 stage was necessary at BOOM's physical dimensions. This confirms that the IPU's buffering approach scales comfortably to production-sized cores. For the single-core configuration, the area overhead including all wiring was 3%. Crucially, this represents an upper-bound: BOOMv2 is substantially smaller than production high-end cores (2.2 mm² compared to 31 mm² for a Zen2 core complex), owing to its smaller BTB, TLBs, absence of SIMD units, lower issue width, and smaller ROB. The IPU's absolute size is constant — the higher percentage overhead reflects the smaller denominator, not a larger IPU. On a Zen2-class core, the same IPU would represent approximately 0.2% area overhead.

VI. IPU CAPABILITIES DEMONSTRATION

Having described the IPU architectural and system design, we now demonstrate the power and generality of this single, programmable architecture. The following subsections are not intended to be a collection of disparate point solutions; we are not claiming to have built the world's best prefetcher emulator or performance monitor. Rather, each example is chosen to illustrate a distinct *class of capability* that the IPU's approach makes possible for the first time in a real-time, in-field setting, *with a single hardware design*. We will demonstrate:

- **Stateful Emulation:** Running complex, history-based algorithms that act at line rate.
- **Software-Defined Performance Attribution:** Replicating and extending the capabilities of specialized, fixed-function analysis hardware entirely in software.

- **Scalable, On-Chip Data Aggregation:** Enabling programmable temporal aggregation, where local, cycle-level analysis allows efficient introspection.
- **Real-Time Component-Level Diagnosis:** Classifying individual microarchitectural events by their internal root cause at the point of occurrence.

Summarized results. Figure 6 shows which hardware signals are connected to the IPU across three of the designs. Figure 7 depicts a utility result for three of the capability demonstrations. An overview of the interface width, code length, and output characteristics is shown in Table IV. For each capability, we first cover its definition, then the demonstration we picked, details of the implementation - which includes the design of the introspection code, the area and power, and whether any data is dropped and its effects on accuracy if so.

Across all four capabilities, the IPU enables runtime, signal-level introspection beyond the reach of fixed-function hardware, tracing tools, or other postmortem analysis.

A. Capability 1: Stateful Emulation

The Capability. The first and most powerful capability we demonstrate is the IPU's ability to execute complex, stateful software in a tight, low-latency situation. This class of task is impossible for existing tools: PMUs are stateless counters, and off-chip trace analysis doesn't run on live deployments.

The Demonstration. To prove this capability, we use the IPU to implement an entangled instruction prefetcher emulator. This task is an ideal illustration, as a prefetcher is a complex, history-based state machine. It must: 1) subscribe to the processor's real-time instruction fetch stream, 2) maintain a complex internal state (e.g., history tables, state transition graphs), and 3) perform computations for every instruction, all without stalling the main processor. The next few paragraphs cover the implementation details.

★ **HIT & Interface.** The HIT is the CPU front end block, and we need essentially one data signal - fetch PC being issued by the processor core (depending on the decoupled front-end design, the signal could be different; a virtual or physical address depending on cache design). The analysis is performed over the entire program so no signal is connected to ADDR nor are TS and TE configured.

★ **Introspection Code.** Implementing the prefetcher using RISC-V code is too slow as it drops data that is required to maintain correct cache state (a state-machine traversal is

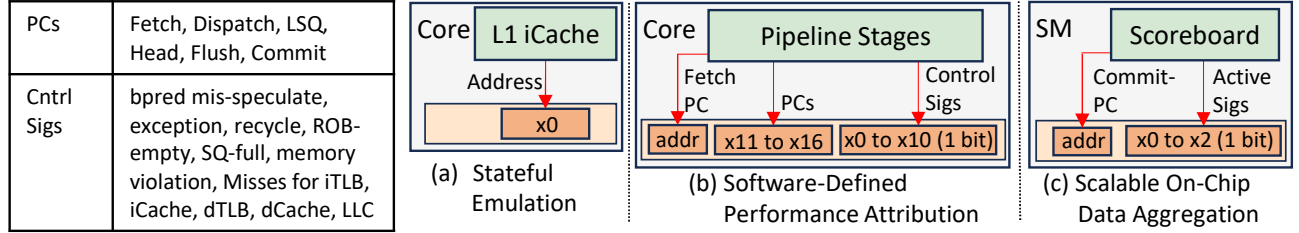


Fig. 6: Demonstration interfaces. (b)PCs and Control Sigs are listed in left-hand table. (c)Active Sigs are for the tensor core, SIMT, and memory subsystem.

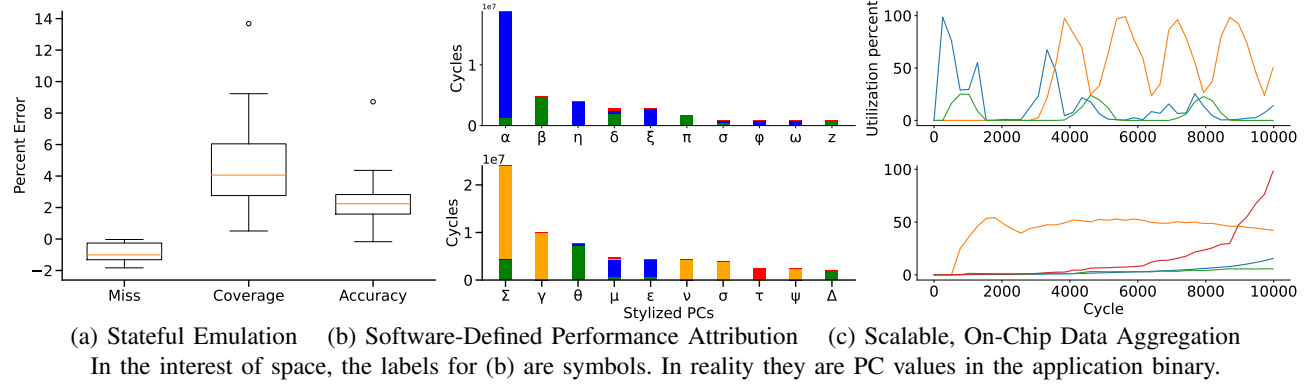


Fig. 7: Utility results for three of the demonstrations. (a) Relative error for each metric in prefetch emulation in-silicon across 135 workload traces. (b) TOP-10 PICS for NAB and astar benchmarks, showing instruction contributions to exposed cycles. ■ Dcache miss, ■ Dcache miss and LLC miss, ■ Drain- SQ full, ■ Misspeculation (c) Cycle-level GPU utilization over time. ■ SIMT, ■ Tensor Core, ■ higher level memory, ■ SIMT. Lower graph sorts running average of statistics when the windows are ordered by overall chip utilization.

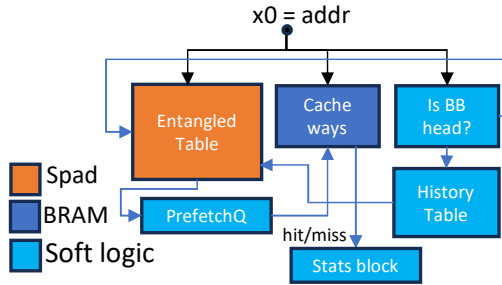


Fig. 8: Organization of the soft-logic block for prefetcher

branch-heavy code). Instead, we implement it on the soft logic and are able to run at one address per cycle with nearly full eFPGA utilization. The RTL design of the prefetch emulator is shown in Figure 8. Since we cannot inject anything into the HIT, our entangling design assumes that all L1 misses are L2 hits to determine entangling pairs (we measure the error this introduces). The execution of memory requests is still correct under this assumption - it does not change the contents of L1.

★ **Performance analysis.** The design accumulates coverage, misses, and accuracy counters in hardware and emits them periodically (every 2^{31} cycles to the host) to avoid overflow.

Thus, the traffic to host is minimal.

★ **Simulation methodology.** In our Champsim testbed we ran 135 CVP traces and compared IPU based prefetch to the original entangled prefetch implementation from the authors. Our results are nearly identical to the results from the original paper, as the only difference is the always-hit-in-L2 assumption, discussed below. To measure error, we compare our statistics to the reference simulation’s statistics.

Analysis of approximations. Figure 7(a) shows coverage, accuracy, and miss-rate error across the traces. For each metric, our always-hit-in-L2 assumption leads to better prefetching than actual, outperforming on each statistic by less than 5% on average. In cases where the prefetch stats are high, the initial miss rate was very low (less than 0.25%) so the other prefetch stats are less meaningful. The Figure shows the distribution of errors in terms in min, max, and inner quartile. Note that we don’t model cache pollution effects, which our results show has small impact on accuracy.

Area and Power. The area of an IPU_{pro} is 0.22 mm². In comparison to the CPU reference, this is 0.7% area overhead; power is 20.8 mW, which is around 0.5% of the CPU reference (Section V). On a chip with a single IPU_{pro} instead of one per core complex, the area and power overheads reduce to 0.175% and 0.125% respectively.

Key Takeaway. We demonstrate that a general-purpose IPU can perform specialized, state-dependent simulation at line rate, in software. This enables inconceivable in-field A/B testing: deploying and profiling prefetcher binaries on live workloads. The contribution is not the prefetcher algorithm, but proof that our general-purpose architecture is fast enough to emulate this complexity—a task previously relegated to offline simulation and impossible on deployed workloads.

B. Capability 2: Software-Defined Performance Attribution

The Capability. The second capability we demonstrate is the IPU’s power to replace rigid, specialized, fixed-function hardware with a flexible, software-defined program. Modern processors often include bespoke hardware blocks for specific, high-value analysis tasks, but these blocks are unchangeable and add to the design validation burden.

The Demonstration. We demonstrate this capability by implementing a “software-defined PICS” (Per Instruction Cycle Stacks). Hardware like TEA [34] is extremely powerful, as it provides developers with PICS: fine-grained, per-instruction attributions for stall cycles (e.g., L2 miss, TLB miss). Its limitation is its inflexibility; it is a custom hardware block designed to find a specific set of predefined bottlenecks. IPU_{lite} constructs these stacks using its programmable core as illustrated in Figure 7b. Implementation details follow next.

★ **HIT & Interface.** The HIT is the core pipeline of a CPU. Figure 6(b) shows the interface as listed in the left table including Control Sigs, which indicate long-latency events starting in the core. In addition, we have 6 virtual address PC values from 4 parts of the pipeline. To restrict program regions, users set the TS and TE registers with the fetch-PC connected to the ADDR register using the IPU API.

★ **Introspection code.** The introspection code has two phases: every cycle we update a Performance Signature Vector (PSV) which is a bit-mask that indicates which event has occurred for a particular dynamic instance of a PC. This follows a sequential if-else-if sequence across all supported hardware events, where if an event occurs in the core pipeline a load-modify-store sequence sets a specific bit of the appropriate PSV to a ‘1’. If an instruction causes a flush, we store its associated PC value in the IPU’s memory, so that we can reference it after it commits. Every 400,000 cycles (TEA paper’s design value) we update PICS which correspond to combining the delays of every dynamic instance of a PC into a single entry. The introspection code scans through the active list of PSVs to determine to which PSV we can attribute cycles and sends to the FIFO that PSV (PC + signature).

★ **Performance analysis.** 215 bits of data are used every cycle. In the common case (representing more than 75% of the cycles - our results and traces we obtained from TEA authors confirm this), no event is triggered when the ROB is sampled (as it isn’t stalled/draind). In the other 25% of cycles, typically a single long latency event occurs. Some times 2 or more events occur when the ROB is sampled (very rarely 3 events, and almost never more than that). Outside of the cycles where the ROB is sampled, there are typically 1-3 events in the

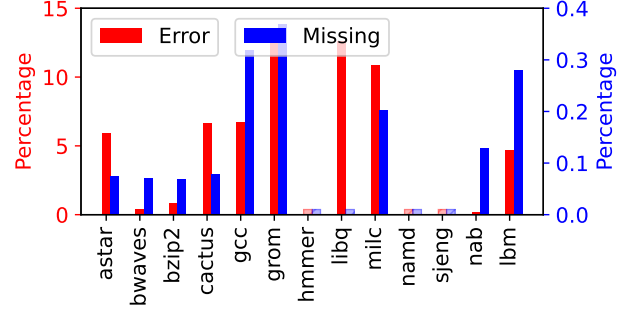


Fig. 9: Average of the relative error for the PICS per benchmark in red and percent of total missed cycles in blue. Light red/blue bars indicate zero error or missed cycles, respectively.

processor pipeline which need to be processed, triggering 3-9 instructions of code. The introspection output data volume is a few bytes of PSV data every 400,000 cycles.

★ **Simulation methodology.** As shown in Table IV, we use a gem5-based simulation. For the SPEC benchmarks, we used SimPoints to simulate representative regions with 100 million instructions per-region. Two example PICS stacks from our 13 applications (all of which we generated the PICS for) are shown in Figure 7(b). For validation of PICS generation, we ran 3 DARCHR microbenchmarks [2] expecting one PC to show up in the PICS stack for these microbenchmarks. The resultant PCs are shown below verifying the generation. One PC has a large cycle-count, showing that is primarily responsible for performance stalls.

PC	Assembly	kCycles	C Code Line
STL2 causes LSQ Full			
4017f6	mov %eax,(%rsi,%rdx,1)	127878	arr[lfsr].p1=lfsr
CCH_st causes Branch Misspeculation			
401813	jne 4017f8	177	if(randArr[i])
ML2 causes D-Cache Miss			
4017ee	mov %eax,(%rsi,%rdx,1)	58595	lfsr = lfsr + arr[lfsr].p1

★ **Analysis of approximations.** This demonstration has the notion of dropped data - if an event is triggered during the PSV generation window of a previous event, we drop that event. Note that IORegs are designed to hold their “old” data (and drop new data) until the IPU reverts back to AP state. To understand the impact of this, we used our simulation testbed to create PICS with simulating introspection code running in 1 cycle vs per-cycle simulation of the introspection (which can take 8 cycles when two events occur in the same cycle). Our error metric is defined as the average relative error (compared to the single-cycle version) of the cycle stack height for each PC for each application. Figure 9 shows this in the red bars. Typically, the quantitative error is < 3% while three applications show 10-14% error. In all scenarios, the list of PCs and the scale of the cycle contributions to PICS was correct, which is most important for performance optimizations. In some very rare scenarios, we drop entire PCs from the PICS stack - when a PC always appears in the

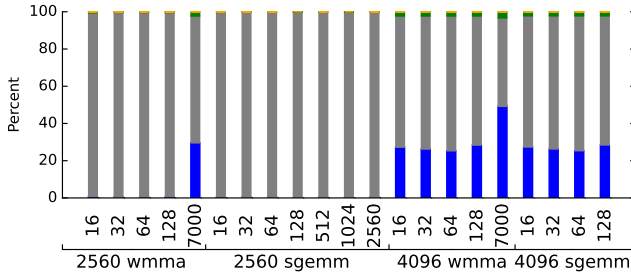


Fig. 10: Breakdown of how use overlaps amongst the three signals collected shown per gemm shape. Blue is all low. Grey is 1 high and 2 low. Green is 2 high and 1 low. The number under each stack is the n and number in groupings is the m and k . wmma or sgemm are the kernels. The leftmost stack is (2560,16,2560) wmma benchmark.

dropped window. The Y-axis shows the percentage of cycles covered by these dropped PC in blue. By definition these are exceedingly rare and unimportant for performance analysis. Across our applications, they cover $\leq 0.37\%$ of cycles.

Area and Power. The IPU_{lite} has an area of 0.019 mm². Compared to the CPU reference, this is an area overhead of 0.06%. Power consumption is 15.0 mW which is 0.38% of the CPU reference power. On a chip with a single IPU_{lite} instead of one per core complex, the area and power overheads reduce to 0.015% and 0.095% respectively.

Key Takeaway. *The IPU successfully generates the same per-instruction breakdown of performance bottlenecks that a dedicated PICS block would. The critical contribution, however, is the demonstration of **generality** for software-defined performance attribution: moving complex performance attribution from a rigid, pre-silicon design choice to a flexible, post-silicon software task.*

C. Capability 3: Scalable, On-Chip Data Aggregation

The Capability. The final capability we demonstrate is the IPU's *programmable temporal aggregation*. This is the ability to process low-level hardware signals every cycle and perform complex analysis on them within flexible, software-defined time windows (e.g., 256 cycles). This class of fine-grained analysis is impossible for existing tools, and is necessary to capture hardware behavior when it is bursty and ephemeral.

The Demonstration. We use the IPU to perform a fine-grained utilization analysis of GPU TensorCores. The introspection program groups activity signals into N -cycle windows, calculating the percentage of active cycles within each window. This allows it to build a histogram, not of average activity, but of the distribution of these fine-grained temporal behaviors. Aggregate PMUs would report a single, misleading average for the entire kernel, hiding the dynamic, sub-kernel-level activity that our IPU's programmable time-windowing reveals.

★ **HIT & Interface.** This is a GPU demonstration, with the HIT for the IPU being an SM scoreboard block. The data signals are 3 one-bit signals indicating whether the SIMT core is active, the TC is active, and the L1 cache subsystem

is in a state where it is servicing one or more outstanding requests (MSHRs non-empty status). While GPU hardware is proprietary the performance counters from NVIDIA Nsight Compute CLI (NCU) count aggregates for these signals indicate they are readily available and not disruptive from a design standpoint. Optionally, the virtual address retiring PC is connected to the ADDR register with IPU's software API used to restrict regions of interest. To isolate to a region within a kernel, a PC start and end range be provided to TS and TE. Or, it can run untriggered to capture the full kernel.

★ **Introspection code.** The introspection code is a single histogram instruction (optimized with a loop directive) that runs for 256 cycles receiving new data every cycle. The output is 3 bytes every 256 cycles, denoting how many active cycles of that unit, which can also be batched across windows.

★ **Performance analysis.** Every 256 cycles, we emit three 1-byte values; thus, introspection output data bandwidth is $3 * 108(\text{\#SMs}) / 256 \text{ bytes per cycle} = 1.7 \text{ GB/second}$ at 1.4 GHz. With 10 of 108 SMs sampled, the bandwidth reduces to 0.16 GB/second.

★ **Simulation methodology.** One representative output is shown in Figure 7(c). The top half shows chronologically ordered windows of 256 cycles with the Y-axis denoting % of cycles in that window where that signal was active. We can see the mutually exclusive behavior. The bottom graph shows the same data ordered by increasing utilization where it plots the running average for each signal. Figure 10 post processes this data and presents it in a different way. We classify windows of 256-cycles into 4 bins: 2 signals high (green), one signals high (blue), all signals low (grey). 'High' means a component is used in greater than 25% the cycles in that window, and 'low' means in less than 25% of the cycles. We can see that large portions of time are spent with at least one component of the SM being idle - pointing to further hardware optimization beyond directions like TMA [56] that have appeared. Other work has also looked at improving such utilization [9], [81].

Analysis of approximations. This demonstration uses our histogram instruction in a novel way, essentially treating each signal as its own bucket and builds a 3-bucket histogram. Hence we have very little dropped data - the source of error is the few cycles needed at the end of a window to write 3 bytes of accumulated statistics.

Area and Power. The IPU_{lite} has an area of 0.019 mm² that is 0.6% of the area of the GPU reference. This demonstration consumes 4.7 mW of power. The IPU_{lite} power overhead is around 0.5% of GPU reference. If instead the chip designer included only one IPU_{lite} on the chip, the area and power overheads are 0.003% and 0.004% respectively.

Key Takeaway. *This “computational introspection” approach effectively breaks the data-scaling barrier. The IPU's ability to be programmed with arbitrary aggregation logic (e.g., “compute histogram,” “find average,” “report max contention”) is the key. It moves the bottleneck from data transfer to local computation, enabling fine-grained, real-time observability.*

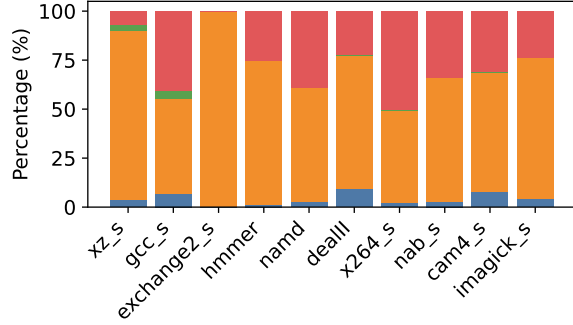


Fig. 11: Each bar represents a breakdown of root-cause classification across all L1D cache misses for a trace. Blue: cold region, Orange: no learned pattern. Green: late prefetch. Red: prefetch failure.

D. Capability 4: Real-Time Component-Level Diagnosis

The Capability. The fourth capability we demonstrate is the IPU’s ability to perform real-time, per-event root-cause diagnosis of microarchitectural component failures. Unlike aggregate counters that report how often a component fails, the IPU classifies why each failure occurred by tapping the component’s internal decision-point signals. The IPU brings this diagnostic capability to **silicon** on deployed workloads without requiring specialized hardware.

The Demonstration. We demonstrate this capability by attaching an IPU_{lite} to the L1 data cache and its prefetcher (we use the Gaze spatial prefetcher [14]). On each demand miss, the IPU classifies the miss into one of four root-cause categories by observing prefetcher-internal decision-point signals. The resulting per-miss, per-PC failure-mode distribution is unavailable from any existing production tool; PMUs report only aggregates, and sampling tools like PEBS lack visibility into prefetcher-internal state.

★ **HIT & Interface.** The HIT is the L1 data cache and prefetcher complex. The data signals consist of the demand miss indicator (1 bit), the Accumulation Table (AT) hit signal (1 bit), the missed_in_pt flag indicating whether the Pattern History Table (PHT) contained a pattern at region graduation time (1 bit), and the MSHR prefetch-in-flight status (1 bit). The demand miss virtual address and PC are delivered via the address bus from the LSQ. To restrict analysis to a region of interest, the user sets the TS and TE registers with the LSQ-PC connected to the ADDR register using the IPU API.

★ **Performance analysis.** The output is a compact histogram of failure-mode counts per PC, emitted periodically to the host. At four categories per PC, the introspection output data volume is a few hundred bytes per reporting interval, imposing negligible host traffic.

★ **Simulation methodology.** We use ChampSim with the Gaze authors’ implementation. We ran 189 Spec benchmark traces and logged the prefetcher’s internal decision-path signals on each demand access. An offline analysis script faithfully models the IPU’s processing latency and classifies each miss,

simulating dropped data that would occur when miss events arrive faster than the 20-cycle processing window. To measure error, we compare the failure-mode distributions with and without the dropped-data constraint. Representative output generated across ten of the traces are shown in Figure 11. For hardware designers, a workload dominated by “no learned pattern” failures points to insufficient PHT capacity or overly conservative training triggers, whereas one dominated by “late prefetch” indicates memory scheduling latency or lookahead distance — two fundamentally different microarchitectural investments that aggregate miss counters cannot distinguish. Programs with a substantial “prefetch failure” fraction, such as x264_s in Figure 11, suggest cache pressure is evicting prefetched data before it is consumed. This is a signal that cache sizing, replacement policy, or prefetch throttling warrant redesign. For software developers, PCs flagged as “cold region” or “no learned pattern” are prime candidates for explicit prefetch hints, since no hardware prefetch is in flight to coalesce with — actionable guidance that no existing production tool provides. Finally, deploying the same introspection binary across a fleet can reveal that the dominant failure mode in production diverges from what design-time benchmarks predicted.

★ **Analysis of approximations.** The primary source of approximation is dropped data: demand misses that arrive within 20 cycles of the previous miss. However, L1D miss events are infrequent relative to the IPU’s processing budget. With the Gaze prefetcher active, the residual miss rate is low, and consecutive misses separated by fewer than 20 cycles are rare. The average error in category proportions per-trace is less than 3.5% with a median of 2.46%, confirming that the processing rate is well matched to the event rate.

★ **Area and Power.** Like the TEA case study, we use an IPU_{lite}, with the analytics code behavior being similar (scratchpad reads and RISC-V instructions), thus resulting in area and power of 0.019 mm² and 15mW.

Key Takeaway. The IPU brings per-miss, per-PC failure-mode diagnosis to deployed silicon, enabling iterative, reprogrammable investigation without a new tape-out. The contribution is not the classification algorithm, but closing the observability gap between simulation and production.

E. Comparison with Existing Mechanisms

Across all four capabilities, the IPU enables runtime, signal-level introspection beyond the reach of fixed-function hardware, tracing tools, or other postmortem analysis. In the first, the IPU emulates a candidate prefetcher via custom logic triggered on memory accesses—something not possible with commercial tools like Intel PCM, CUPTI, or CoreSight, which lack in-field programmable logic. While LBA [12], [13] is conceptually similar, it lacks the programmability and hardware efficiency of IPU. In the TEA case, performance counters cannot implement Time-Proportional Event Analysis (TEA) [34], which underpins PICS. Unlike prior work requiring dedicated RTL for RISC-V BOOM, our IPU replicates TEA without RTL changes and with greater flexibility. The

Yr	Technique	S	P	T	A
General Approaches					
–	SW profiling	N	N	N	Y
–	Perf counters	Y	N	p	Y
–	Debug monitors	Y	N	Y	N
–	HW emulation	N	Y	Y	N
–	RTL simulation	N	Y	Y	N
Security					
'03	DISE [17]	N	p	N	N
'10	FlexCore [25]	Y	p	Y	N
'11	LBA [12], [13]	N	Y	N	N
'20	PHMon/Nile [23], [24]	Y	p	N	N
Embedded Systems					
'10 ¹	ABACUS ³ [27], [48], [69]	Y	p	N	Y
'13	hidICE Verification [7]	Y	Y	N	N
'15	SOF [41], [42]	Y	N	p	N
'16 ¹	AIPHS ² [51], [73], [74]	Y	N	p	Y
'17	Enhanced PMU ² [66]	Y	N	p	Y
'18	NIRM [68]	Y	N	N	N
Profiling and PerfMonitors					
'01	Programmable Co-Proc [52], [82]	Y	Y	p	N
'01	Stratified Sampling [65]	N	Y	p	N
'03	ULF [80]	p	Y	N	N
'03	Interval Based Profiling [53]	N	Y	p	N
'05	Owl [67]	N	p	Y	Y
'25	IPU (ours)	Y	Y	Y	Y

¹Year of most relevant work ²Limited to manipulating event counts

³Limited to architectural traces. *p* means partial.

TABLE V: Related work in our 4-axes taxonomy. S (Speed); P (Programmability); A (Accessibility); T (HW Transparency)

IPU can also build histograms of fine-grained, cycle-level signal states, which are impractical to obtain via simulation, emulation, or debug monitors due to visibility limits or high overheads. Finally, the IPU classifies each demand miss by its prefetcher-internal root cause—a per-event, per-PC diagnosis that neither PMUs nor sampling tools can provide.

VII. RELATED WORK

We define a four-attribute taxonomy to classify hardware introspection (Table V): **Speed** (line-rate, non-disruptive execution); **Transparency** (visibility, from coarse counters to fine-grained microarchitectural behavior); **Programmability** (flexible, post-fabrication analysis, not fixed-function queries). Finally, **Accessibility** measures how readily a solution can be used by both hardware and software developers, especially for in-field deployment. It gauges the ability to gain actionable insights without being obstructed by hardware constraints or proprietary software bottlenecks, ensuring the technique is broadly and practically available to all stakeholders. Prior works' trade-offs create a capability gap, which the IPU fills.

Software profiling tools (*strace*, *gprof*, binary instrumentation) lack hardware transparency and scale poorly with introspection complexity. *Performance counters* offer speed but are restricted to pre-defined events with no support for stateful emulation (Table VI). PGO techniques [10], [11], [22], [35], [43], [45], [50], [54], [61], [62], [78], Top Down Analysis [76], [79], and Intel PT [36] improve optimization but address only part of the opacity problem. *Debug monitors* [37] (JTAG, boundary-scan) offer deeper access but require physical connections and cannot efficiently capture live

Feature	CoreSight	Intel PT/PMU	IPU
Granularity	Trace-level	Event/sample	Per-signal
Programmable	No	Configurable	Full
Control interface	Hardwired	MSRs	API

TABLE VI: IPU vs. existing introspection mechanisms

execution. *HW emulation* (Cadence Palladium [59], Synopsys Zebu [60]) provides full transparency but costs millions and runs at $2\text{--}5\times$ slowdown; RTL and cycle-level simulation are orders-of-magnitude slower still.

Prior academic work falls into three categories, none achieving the IPU's combination of speed, programmability, transparency, and accessibility. Security-oriented approaches (DISE [17], FlexCore [25], LBA [12], [13], PHMon/Nile [23], [24]) run at-speed but eschew programmability and transparency. Embedded systems techniques (ABACUS [27], [48], [69], hidICE [7], SOF [41], [42], AIPHS [51], [73], [74], Enhanced PMU [66], NIRM [68]) assume direct hardware access, lacking accessibility and programmability. Performance counter and profiling techniques [47], [53], [65], [67], [77], [80] lack hardware transparency. Owl [67] proposed FPGA monitoring but was limited to pre-defined analytics, while Profiling Co-Processors [52], [82] lacked fine-grained transparency beyond instruction retirement signals. Our work builds upon these ideas, leveraging the RISC-V ecosystem, modern accelerator design [30], [72], and specialization principles [20], [30], [55] to deliver a hybrid architecture that spans all four axes.

VIII. CONCLUSION

This paper introduced the **Introspection Processing Unit (IPU)**, a new architectural primitive designed to solve a fundamental "capability gap" in hardware observability—the gap between inflexible, low-overhead PMUs and high-overhead, post-mortem trace solutions. Our core contribution is the IPU's novel hybrid microarchitecture, which pairs a programmable RISC-V core with judiciously chosen extensions and small hardware accelerators. We demonstrated this architecture's power not by solving individual problems, but by showing it can perform entire *classes of analysis* previously intractable in the field, including stateful emulation, software-defined performance attribution, scalable on-chip data aggregation, and real-time component-level diagnosis. Our complete RTL-level prototype proves this new capability is practical, adding $< 1\%$ in area and consuming $< 25mW$.

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